Self-Gravity with Mesh Refinement in the AthenaK Shearing Box

Program Summary: Phases 0–3 — Collapse, and the Stratified Gravito-Turbulent Box

`athenak-multigrid` tree, branch `multigrid`

August 25, 2026

Goal and route

Target science: shearing-box simulations of gas + dust with self-gravity and AMR zoom-in on gravitationally induced dust clumps (streaming-instability / GI-arm interaction, cf. Baehr, Zhu & Yang 2022), and the Gammie (2001) gravitoturbulence program. Chosen route: a **multigrid Poisson solver with shear-periodic boundary conditions on refined meshes**, with a uniform-grid **FFT solver as the machine-precision oracle** throughout. All work is validated against analytic solutions, the FFT oracle, and the published Athena++ results the solver descends from (Tomida & Stone 2023). Full details: `multigrid_selfgravity_validation.pdf` (32 pp.), reproduced end-to-end by `multigrid_validation_tests.ipynb` (44 code cells); the FFT track has its own companion document.

Phase 0 — shearing box + mesh refinement infrastructure (hydro)

- **Orbital-advection toggle**: a fully consistent non-FARGO mode (measured cost 5.7×: CFL against the full shear flow), enabling A/B tests and future patch refinement.
- **Refinement policies**: interior-x_1 / uniform-boundary-level rule (both shear faces share one level), and the *annular* policy under FARGO — refined regions span complete azimuthal rings, so the per-level y-shift only ever involves same-size neighbors.
- **AMR**: shear-boundary communication state rebuilt after every remesh; a graded-refinement cap keeps 2:1 balancing off the boundary columns; an upstream derefinement-ordering bug (partial ring splits) found and fixed.
- **Validation**: ring SMR bit-identical to reference runs; 20-orbit epicycle endurance — amplitude drift 0.008% (ring+FARGO) vs. 0.064% (uniform); AMR with a prescribed moving ring: 2296 blocks created / 2240 destroyed over 20 orbits, mass conserved exactly, matches the static ring an order of magnitude closer than either matches the uniform grid.

Phase 1 — shear-periodic multigrid boundary conditions (uniform)

Shear-periodicity enters as a boundary *identification* only: a y-remapped x_1 ghost fill (conservative PPMX remap) applied at every level of the multigrid hierarchy through global boundary planes; the 7-point operator is untouched. Key results: solves at shear-periodic instants are **bit-identical** to plain-periodic solves; the analytic shearing-wave potential is recovered *below* the FFT solver's own error (9.5×10^{-8} vs. 1.6×10^{-6} at 128^3); the swing-amplification history matches FFT to 2×10^{-7};

V-cycle contraction degrades only 0.116 → 0.157 under a near-worst shear phase. Anchored to Tomida & Stone (2023): their §4.1 convergence factor (0.125), second-order accuracy (identical to FFT to all digits), and §4.3 Jeans-wave results all reproduce.

Phase 2 — multigrid gravity on refined shearing-box meshes

phase	configuration	headline validation
2a	static interior rings (boundary blocks at root)	guard-only change; ring solves bit-identical to periodic twins at qomt = 0; L_2 2nd order; swing peak shifts *toward* theory, gravity-off control dominates ⇒ solver adds no error
2b	boundary annuli (refined blocks *on* the shear faces)	sheared octet ghost fill (host-side, rank-replicated, no MPI seam); bit-identity at 1 and 2 levels; contraction unchanged (0.18–0.20)
2c	AMR	shear planes/offsets/octet flags rebuilt per remesh; moving-ring swing: several hundred remeshes, no solver warnings; np=2 ≡ serial to every printed digit

The octet (refinement) machinery inherited with AthenaK #776 was first anchored on the isolated binary-potential test (T&S §4.2): region-wise second order against the analytic two-sphere solution. Throughout Phase 2 the refinement interface — not the shear — owns the error budget, and every dynamic test carries a gravity-off control proving the gravity solver contributes nothing beyond hydro truncation. One practical finding: refined meshes should run FMG with a *fixed* V-cycle count (T&S practice); the stagnation-exit stopping rule can hit its iteration cap mid-run on wound-up shear modes.

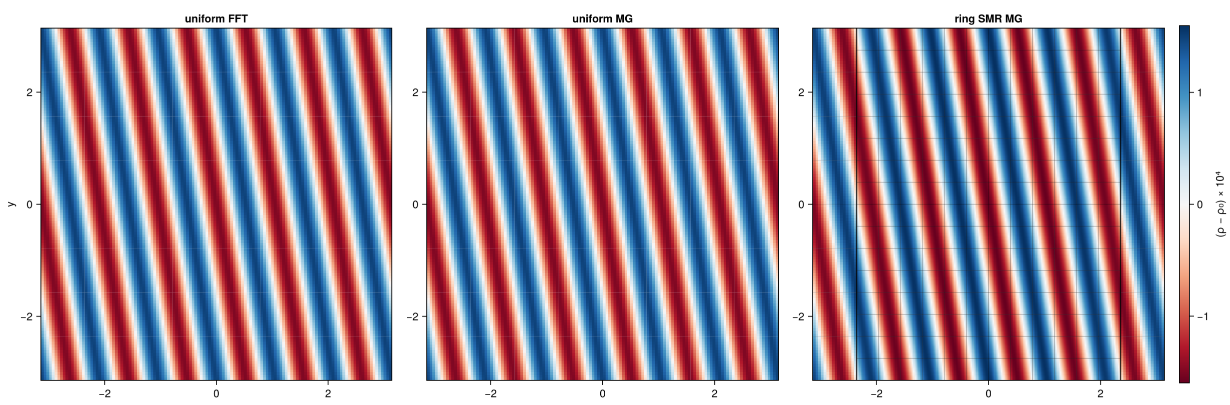

Figure 1: The Phase-2a swing test: final density for uniform FFT, uniform multigrid, and ring-SMR multigrid — the two uniform panels agree at the float32 quantization floor, and the wave crosses every refinement boundary without artifact.

Into the unstable regime: first collapse (this week)

With the machinery validated, the swing test was pushed past the box-scale Jeans threshold ($4\pi G > \kappa^2 + k^2 c_s^2 = 2$). An empirical ladder established what clumping actually requires: transient waves *bounce* unless the compressed crest itself becomes Jeans-unstable during the unstable window ($\lambda_J(\rho_{\mathrm{crest}})$ below the crest width). At $4\pi G = 6$, amp $= 0.2$ the crest fragments at $t \approx 4.5$ and collapses into a **bound clump at** $\rho = 203\rho_0$ — beyond the uniform-grid Truelove ceiling ($171\rho_0$), motivating the AMR zoom, which follows the crest and clump with density-triggered rings (the annular policy answers "how do you zoom under FARGO": rings, completed automatically by the flag-sync, not local patches). The multigrid solver ran the entire collapse — a 200:1 density contrast on a shear-periodic adaptive mesh — with zero convergence failures.

The program is now **deployed on NAS HECC**: a 256^2 discriminator run at 64 ranks confirmed the two-site fragmentation as physical (the 128^2 single clump was an under-resolution artifact), and its t=14 extension followed the winner-takes-all endpoint — runaway to $2.6 \times 10^5\rho_0$ with the companion consumed — with the solver machine-clean through the quarter-million contrast. The deployment also exposed (deterministically, at the same timestep in two attempts) a latent **upstream bug** in the shearing-box x_1 exchange: sender and receiver computed the integer shear shift from their own MeshBlocks' stored cell sizes, whose last-ulp roundoff can disagree near integer crossings, desynchronizing MPI message sizes. All seven shift computations now derive from bitwise-identical global mesh quantities; the fix is validated from serial (bit-identical to prior references) to 64 ranks.

Finally, an **FFT-solver cross-check** closed the last audit channel: a serial FFT twin of the 256^2 collapse matches the multigrid run to 5×10^{-8} through the linear phase and reproduces the *same two fragments within a third of a cell* after five nonlinear dynamical times; extended to t=14, both solvers reach the same fate on the same clock (ρ=100 crossings within 0.05), diverging only *beyond* the Truelove ceiling (FFT saturates near it; multigrid runs away unresolved) — a solver-based demonstration that past-ceiling densities are resolution artifacts. The y-boundary fragmentation has now survived four independent audits: resolution, mesh topology, rank count, and solver identity.

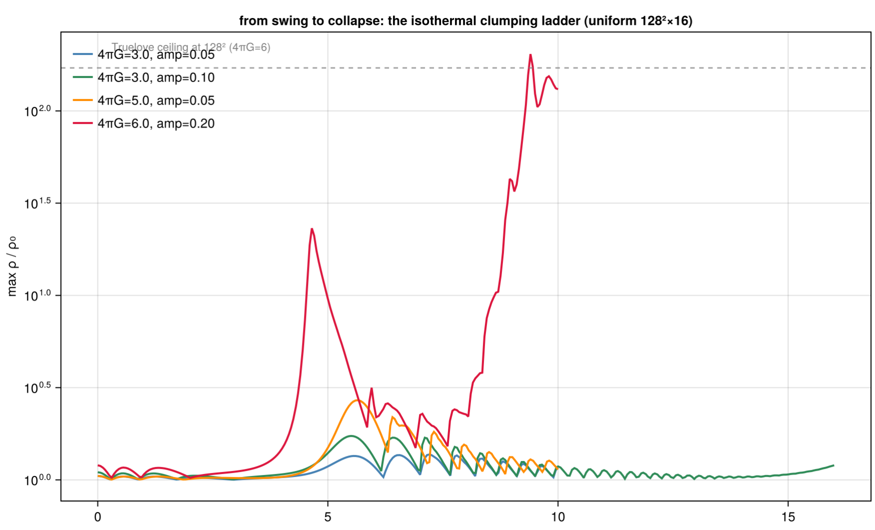

Figure 2: The nonlinear ladder: maximum density vs. time for the four uniform-grid runs, with the Truelove ceiling marked.

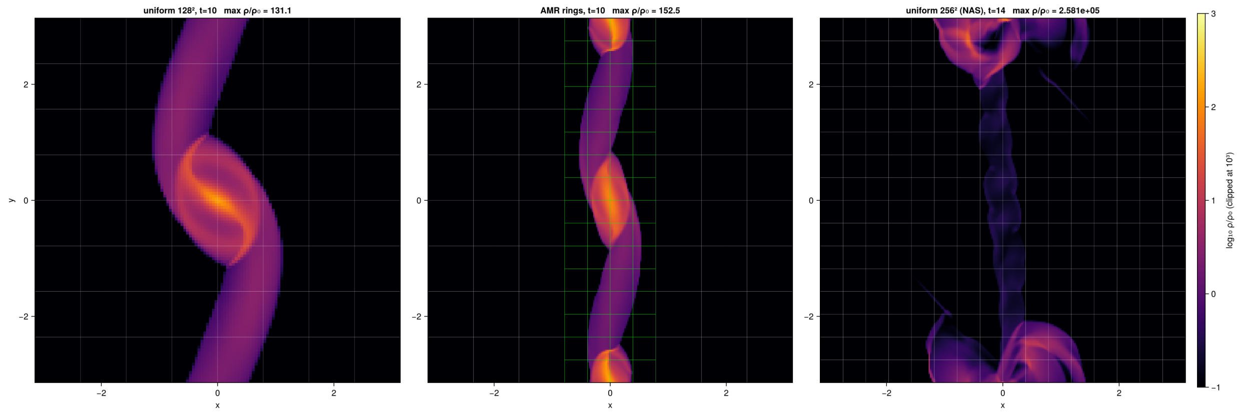

Figure 3: Final states of the collapse runs ($\log_{10}\rho$, MeshBlocks drawn, green = level 1): uniform 128^2 (fragments merged by under-resolution), the AMR run resolving both clumps, and the 256^2 NAS run at $t = 14$ in the winner-takes-all runaway.

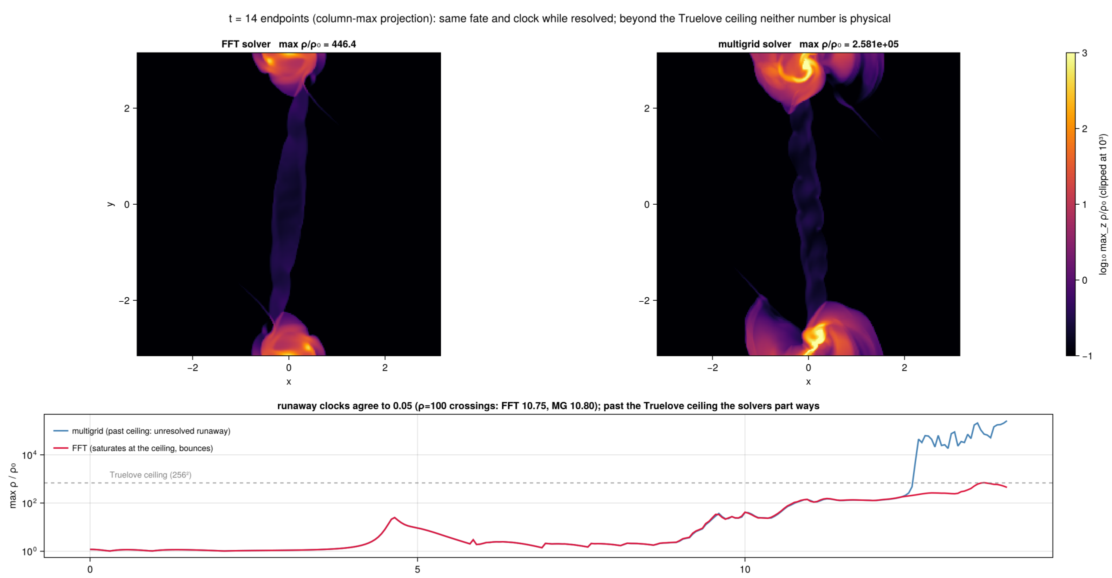

Figure 4: The solver cross-check at $t = 14$ (column-maximum density projections): FFT and multigrid agree on fate and clock while the collapse is resolved, and part ways only beyond the Truelove ceiling (dashed) — neither number is physical there.

Phase 3 — the stratified box: slab-open gravity, cooling, and gravito-turbulence (this week)

The third dimension of the physical problem. Three deliverables:

(i) Open (vacuum) vertical boundaries for the multigrid solver (`mg_bc = slab`): each solve builds exact Dirichlet face planes from the *discrete* infinite-vacuum Green's function of the 7-point stencil (μ-decay per horizontal mode at the shear-shifted k'_x; the discrete slab potential for the now-physical mass sector), applied at every multigrid level. The converged interior equals the infinite-domain discrete solution *exactly*: a new closed-form "sheet" arbiter validates this absolutely (no mean subtraction) at 1.2×10^{-12}, against the independent FFT-KO09 method at 3×10^{-13}, bit-identically across MPI ranks, and at 8×10^{-5} through five dynamical times of *nonlinear evolution*. Shear, SMR, and dynamic AMR all compose with it; the only refinement cost is a constant *force-free* $O(h^2)$ offset in enclosed refined regions — the dipole signature of the conservative level interface, invisible to every periodic-box test that came before.

(ii) Stratified-box physics: the vertical tidal force's missing energy work term (a latent upstream gap, invisible to isothermal runs), SC14's two cooling laws with timestep guards, and a boundary lesson — plain `outflow` z-boundaries feed a gravity-driven *inflow runaway* (halo mass up 100×, disk destroyed by $t \sim 8\,\Omega^{-1}$); the `diode` boundary makes the SC14 initial state hold hydrostatically to 2% of c_s. AMR refinement is automatically vetoed on the x_3 faces, and a clump-triggered AMR run tracked collapse clumps through refine/derefine cycles with the slab planes rebuilt at every remesh.

(iii) The Shi & Chiang (2014) reproduction: a `gravito_turb` pgen integrating their self-gravitating vertical equilibrium (their published constants reproduce the Q_0=1, H=1 calibration to 5×10^{-4} — an independent audit of the paper), with their Q, α (eq. 19), α' (eq. 20) diagnostics as history outputs. The half-width production run ($[-16,16]^2\times[-6,6]H$ at their resolution, $\beta = 10$, 4 MPI ranks, 6.8 h on a laptop) reproduces their Figure 2 end to end (Fig. 5): cooling collapse, vertical rebound, then $250\,\Omega^{-1}$ of self-regulated gravito-turbulence with, over their averaging window, $\langle c_s\rangle_\rho = 2.187$ (theirs: 2.18), rms $\delta v = 1.68\,H\Omega$ (1.79), $M/M_0 = 0.92$ (~ 0.9), and $\alpha_{\rm grav} = 0.034$ (0.039). The residual gaps ($Q = 1.11$ vs 1.33; a burst-inflated Reynolds channel) all carry the signature of a single box-scale swing mode recharging quasi-periodically — the half-width-box effect — which the **full-width $64H$ run now queued on NAS** will resolve; the β-scan (their Table 1, via their "morphing" restart chain) is packaged alongside it (`scripts/cluster/`).

Status and next steps

Delivered: FFT solver (validated, incl. open-z, and a full nonlinear cross-check twin); shearing-box refinement infrastructure incl. AMR; multigrid gravity on uniform, SMR-ring, boundary-annuli, and adaptive meshes, serial and MPI, now with slab-open vertical boundaries for stratified disks; cooling; the SC14 configuration, analysis tooling, and cluster kit; ∼40 pages of validation documentation, the reproducing notebook, and versioned test inputs for every configuration. **Next**: (i) the full-size SC14 runs on NAS (β-scan, resolution doubling, and AMR zoom-in on gravito-turbulent clumps — the configuration no FFT code can run); (ii) dust coupling (athenak-dust PM particles + IMEX drag) toward the SI+GI clumping science (Phases 4–5); (iii) deferred infrastructure: host-root shear fill, MHD face-centered remap with refinement, and (if deep zooms demand it) per-ring bookkeeping for patch refinement under FARGO.

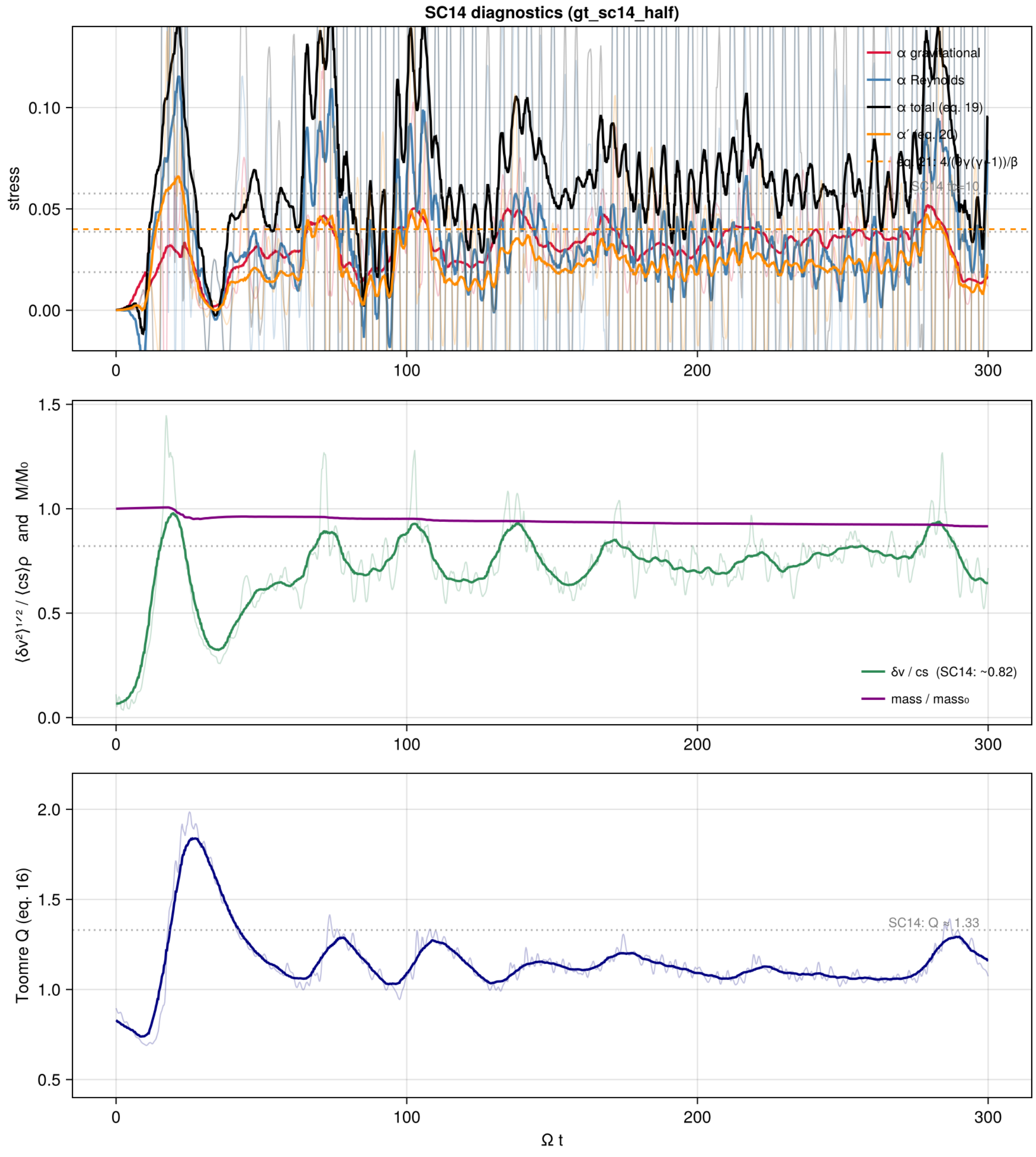

Figure 5: The SC14 $\beta = 10$ half-box run in their Figure-2 form (thick: $12\,\Omega^{-1}$ boxcar; dotted: SC14's tc=10 values; dashed: their eq.-21 energy-balance line): stresses, $\delta v/c_s$ and mass, Toomre Q.

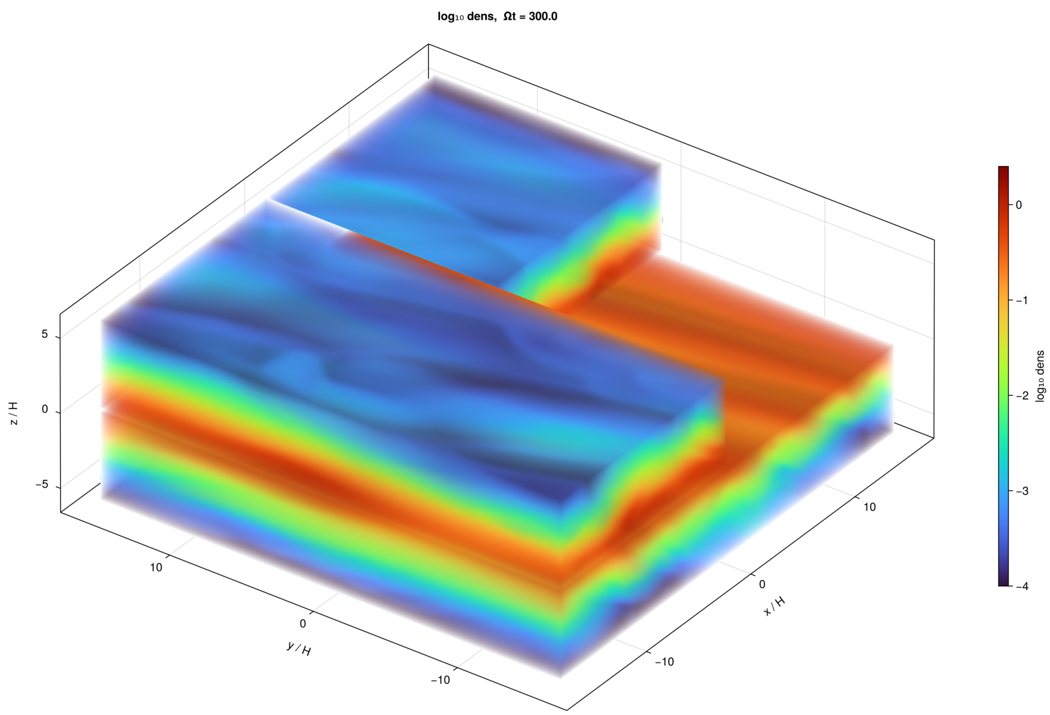

Figure 6: $\log_{10}\rho$ at $\Omega t = 300$, cutaway volume rendering in the style of SC14's Fig. 2: gravito-turbulent midplane wakes under the halo they force — the configuration (stratified, open-z, refinable) that no FFT solver can host.